



Patterns, so predictable

Pattern locks have now become as common and predictable as passwords. Read the entire study here. <http://dgit.in/cmnptrns>



Why Microsoft, why?

Microsoft has no intention of telling us what's within Windows 10 patches. Read more here: <http://dgit.in/winxpctch>

Military tech

landing manoeuvres. The F-35C is a larger version featuring increased wingspan and reinforced fuselage to aid in carrier-based takeoff and landing.

Jack of All Trades

The uniqueness of the F-35 programme can't be fully understood without a primer on different types of military jet aircraft. This includes fighter jets which are primarily focused on interdiction – that is engaging and defeating enemy jet aircraft. This type of military jets are designed with an emphasis on manoeuvrability, speed and a blend of armament as well as avionics for hunting down and destroying enemy aircraft. The F-15 Eagle is a good example of a dedicated fighter jet. Bombers such as the B-2 Spirit eschew manoeuvrability and speed for increased range and payload capacity in order to reach out and destroy ground-based targets. These aircraft have a significantly higher operational ceiling to escape ground-based defenses, but they are still vulnerable to enemy interdiction. Attack jets combine the best virtues of fighters and bombers in a platform designed to enter hostile territory and take out enemy assets with pinpoint precision and great speed. These aircraft may not have an impressive payload capacity or range, but that also enables them to be fast and able to defend themselves against enemy aircraft. USAF's trusted tankbuster A-10 Thunderbolt II is the finest example of a military attack jet.

Aircraft that combine some or all of these attributes are classified as multirole fighter jets. There have been excellent multirole fighters in the USAF arsenal such as the F-16 Fighting Falcon and F-18 Hornet, but the F-35 programme takes the concept of a multirole fighter to the next logical level and includes the vertical landing and hovering capability of the AV-8B Harrier II. Developed jointly with UK, the Harrier II is a one-of-kind ground attack aircraft that's capable of vertical takeoff and landing – a feature that isn't found in any other military jet due to the monumental engineering complexity and limitations involved. The F-35B is the first multirole fighter to incorporate Short Takeoff and Vertical landing (STOVL). It

achieves this with a rear jet exhaust that can be swivelled ninety degrees to create pure upward thrust. This works along with a massive centrally-located lift fan capable of generating enough upward thrust to complement the downward pointing rear nozzle. These two generate enough lift to enable short takeoff, hovering, and vertical landing.

Master of None

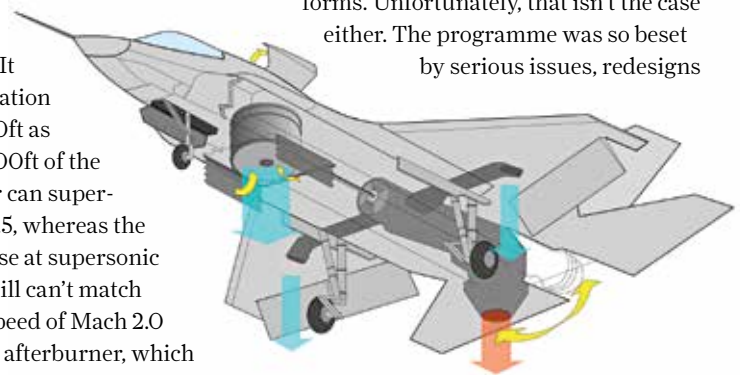
This unique capability stems from the US Marine Corps requirement of a close support aircraft that's capable of being deployed and land even in conditions where there's no access to a conventional airstrip. This essentially makes the F-35 the Swiss Army knife of military aircraft. Unfortunately, the problem with the Swiss Army knife is that it isn't as good at individual tasks as the tools specifically designed for those tasks. This phenomenon is reflected in the F-35 as well. It has a lower operation ceiling of 48,000ft as opposed to 65,000ft of the F-22. The Raptor can supercruise at Mach 1.5, whereas the F-35 cannot cruise at supersonic speeds. And it still can't match the F-22's max speed of Mach 2.0 despite using an afterburner, which can propel the F-35 to a maximum speed of just Mach 1.6. The extra weight added by the STOVL capability severely restricts the combat radius of the F-35B down to 450 nautical miles as opposed to the 600 nautical miles figure of the F-35C.

The worst victim of this engineering complexity is manoeuvrability. The F-16, a model that is supposed to be replaced by the F-35, is capable of taking sustained turn load of 9g, whereas the F-35 had its maximum sustained g loading dropped down to 4.6g. It's also objectively slower than the F-16; taking 43 seconds longer than the outgoing multirole jet to accelerate from Mach 0.8 to Mach 1.2. What's more, the F-35 also requires complex manoeuvres to reach its top speed of Mach 1.6, but doing so isn't practical since it uses up almost all of its onboard fuel. Needless to say, according to a leaked

report on a mock dogfight conducted between the F-35 and F-16, the former didn't have a chance against its faster and more manoeuvrable predecessor in a mock dogfight between the two. Worse yet, the F-16D it engaged unsuccessfully in the mock dogfight was bogged down with the added weight and aerodynamic drag of external fuel tanks. This report's authenticity was later confirmed by F-35 maker Lockheed Martin itself. On the other hand, the US government-funded global military think-tank, RAND, was a lot more damning in its report, which said that the F-35A "can't turn, can't climb, can't run" while describing its air-to-air combat performance.

A Trillion Dollar Mistake

Well, you might argue that at least the F-35 is supposed to be a cost effective replacement to multiple fighter platforms. Unfortunately, that isn't the case either. The programme was so beset by serious issues, redesigns



The F-35 hovers using a swivelling rear exhaust and central lift fan

and delays that the unit procurement cost of each F-35 is upwards of \$300 million and nearing that of the super expensive, but also highly competent F-22 Raptor. The F-35, on the other hand, can't even outfight outdated fighter aircraft it was designed to replace. Then there are other serious unresolved issues such as fire hazards posed by the fuel storage design, serious wing drop problem, poor radar function, ejection seat failures and aircraft software that has been a constant source of complaints. With a total development budget of \$1.5 trillion and counting, the F-35 programme is turning out to be one of the most expensive mistakes committed by the US armed forces. **[1]**